Requirements Engineering (II)

Training for the IREB CPRE
Foundation Level test

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1h1/4

EU 2 – SYSTEM AND SYSTEM CONTEXT

At the end of this chapter, you should

1 Know system context, system boundary and context boundary

2 Master and use system boundary and context boundary

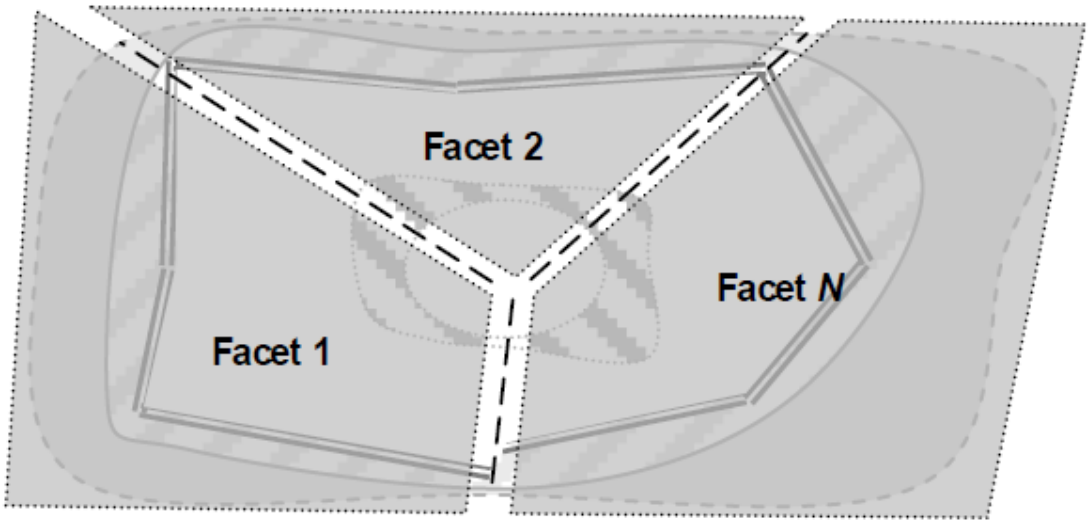
The system context

- A system is always part of an environment
- The **system environment** influences the system requirements in a significant way
- The **system context** is the relevant part of the environment
- Requirements only exist in a context

[Pohl 2010] « Requirements Engineering » §5.1

Principles of structuration of the context

1. Devide the context into facets

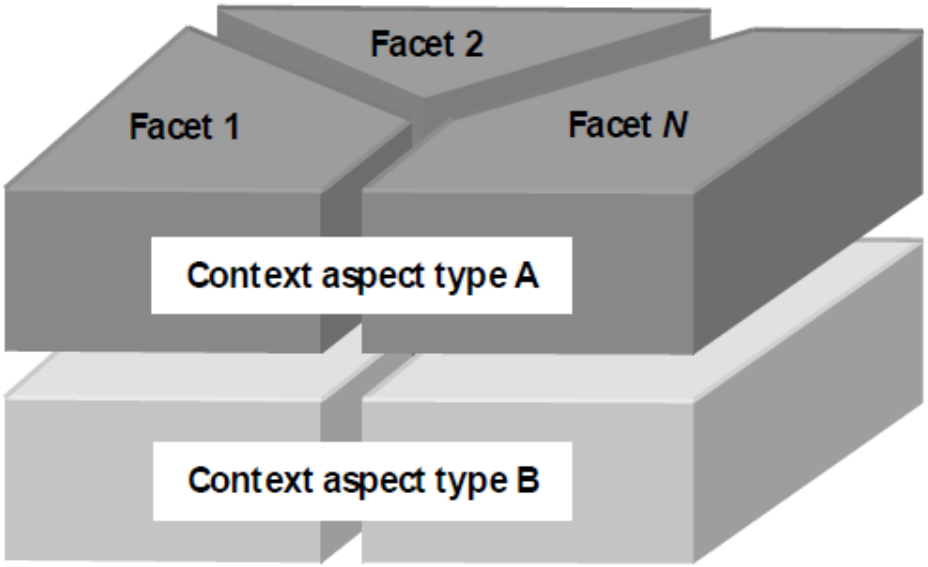


Stakhlders can focus on facets at any time

[Pohl 2010] « Requirements Engineering » Fig. 6.1

Les principes de structuration du contexte :

2. Classify aspects within facets



Alllows to specify each context in a systematic way

[Pohl 2010] « Requirements Engineering » Fig. 6.2

The 4 facets of the context

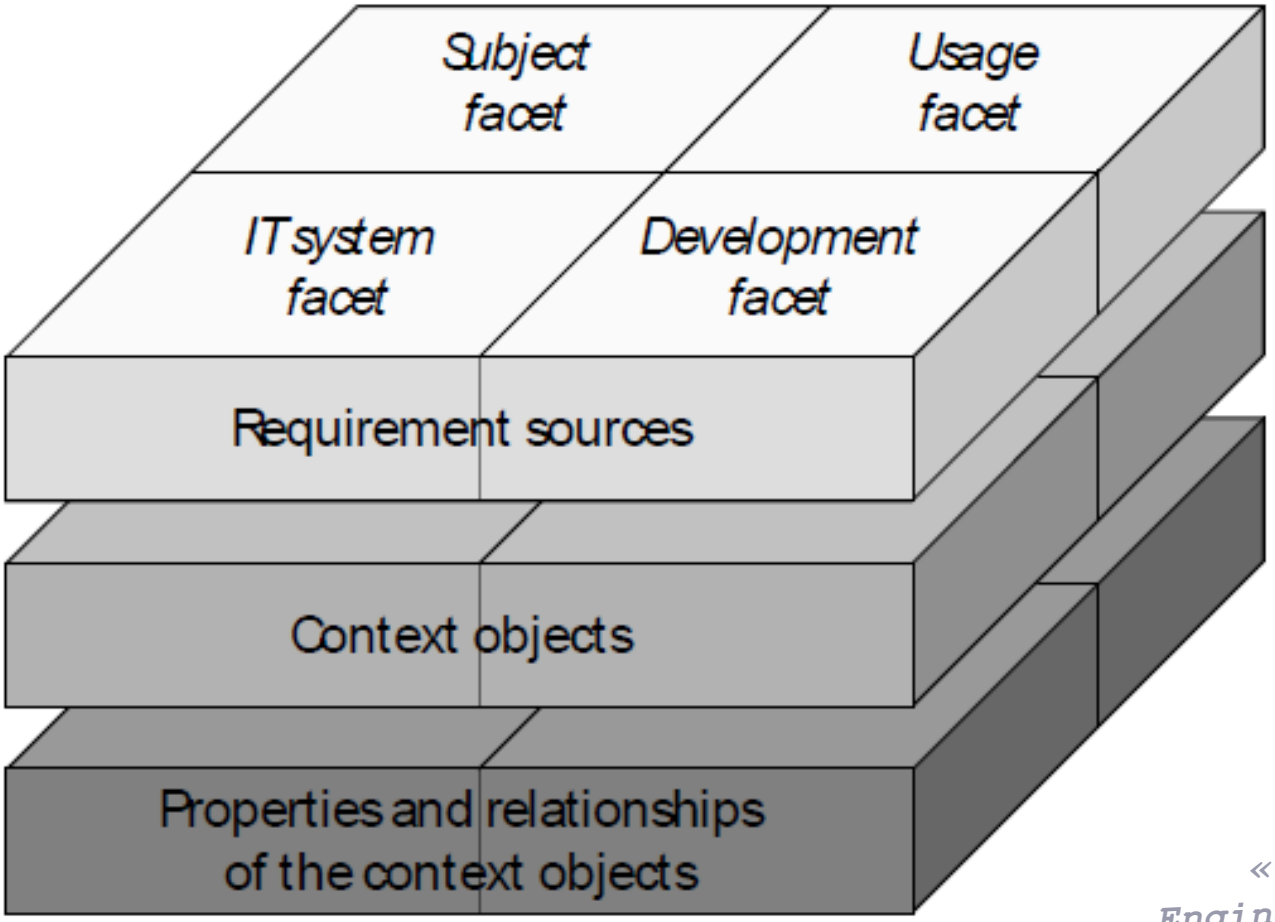
- **Subject**
  Objects and events that need to be represented in the system
- **Usage**
  how people and other systems use the system
- **IT**
  operating/technical environment, physical and software components, IS processes and strategies
- **Development**
  all aspects that influence development procedures

3 aspects in each facet:

[Pohl 2010] « Requirements Engineering » §6.2

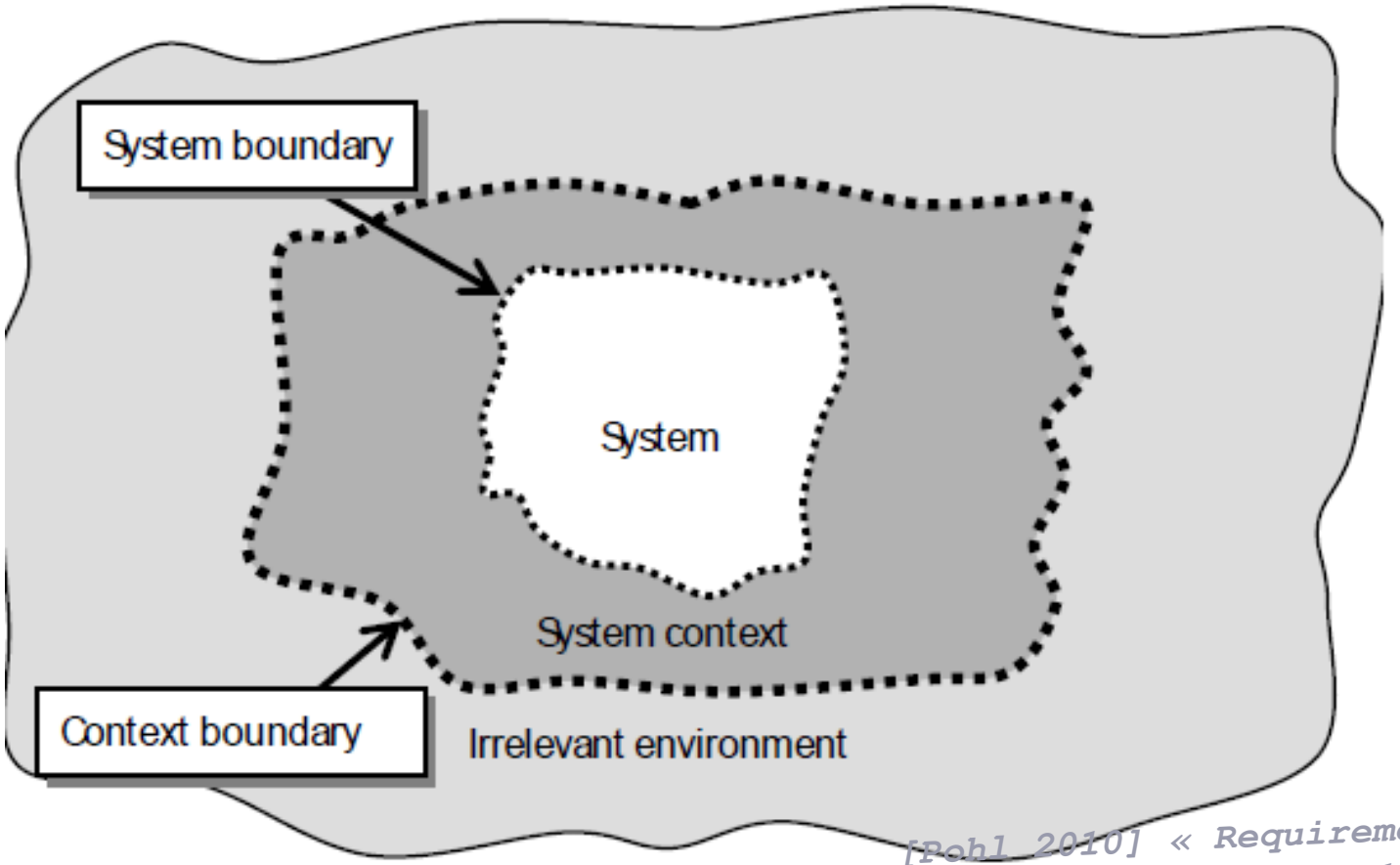
- Requirements **sources**:
  - stakeholders, documents, other systems
- **Objects** of the context:
  - People and artefacts that should be considereddduring RE
- **Properties and relationships**:
  - How things and poeple in the context relate to each other

4 facets, 3 aspects



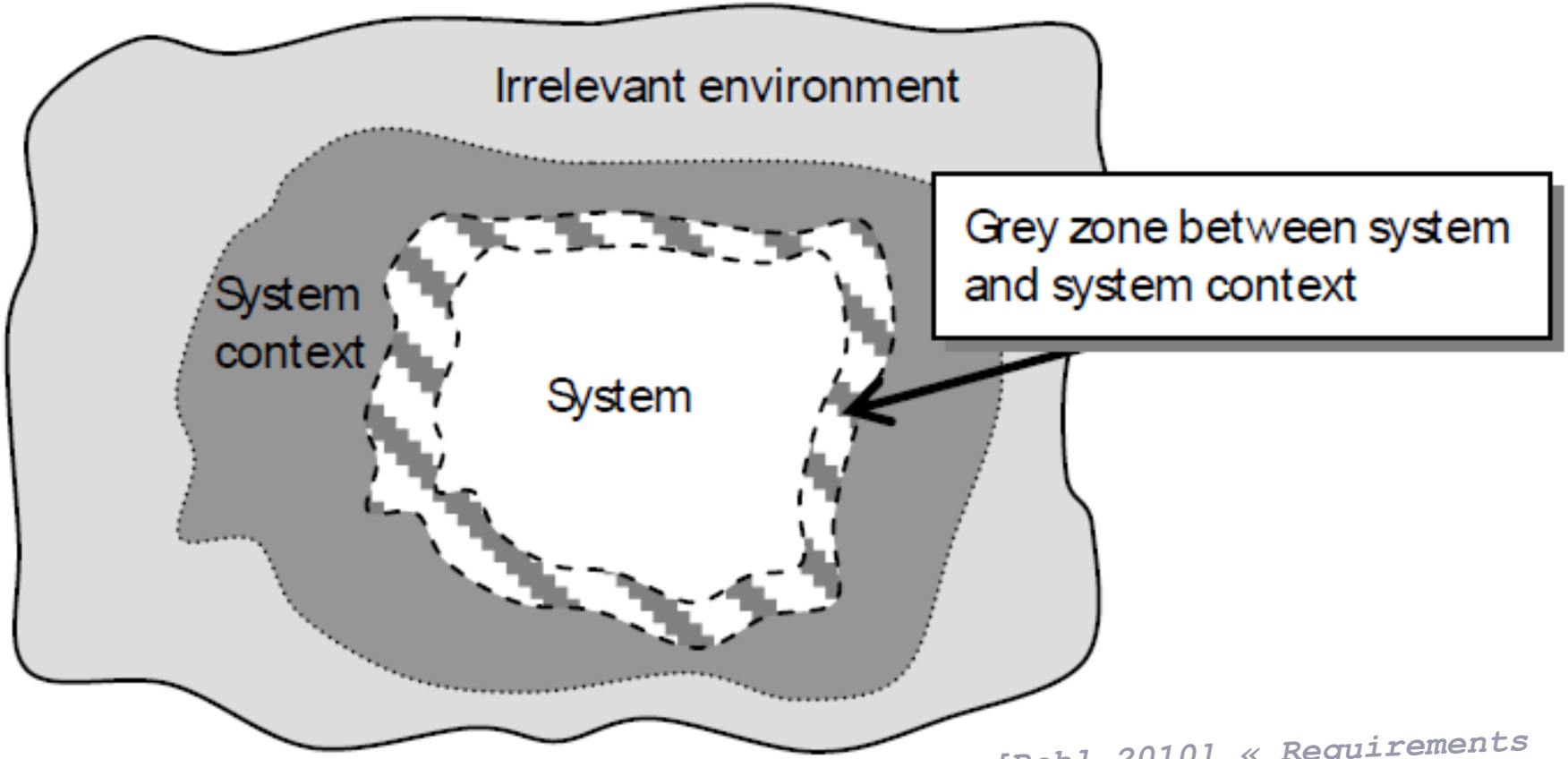
[Pohl 2010] « Requirements Engineering » Fig. 6.3

Boundaries



[Pohl 2010] « Requirements Engineering » §5.1

The grey zone

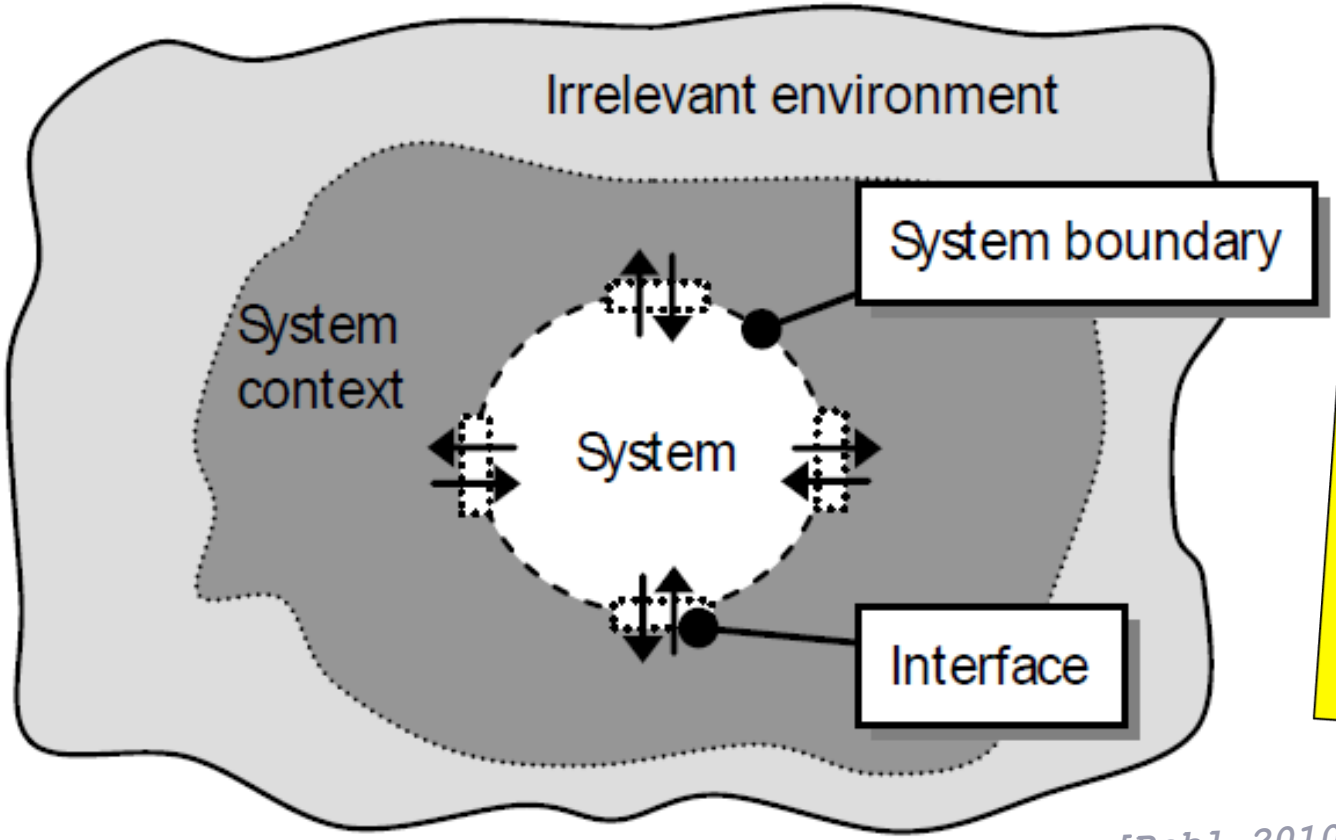


[Pohl 2010] « Requirements Engineering » §5.2

System boundary

- Separates modifiable and stable artefacts
- Initially not known with precision (« grey zone »)
- Adjustments will be needed

System interface

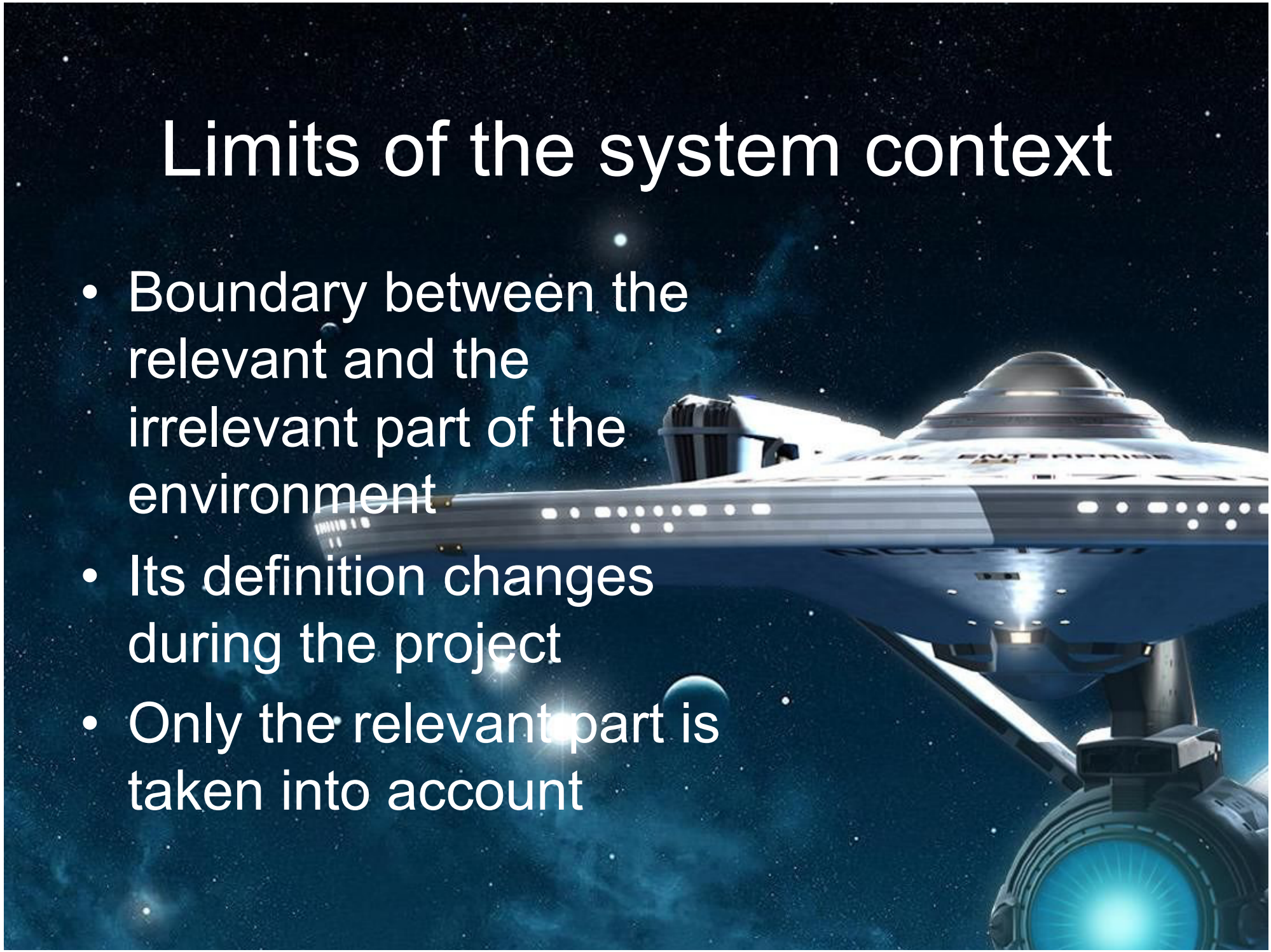


Interface : the interaction channel between the system and its environment

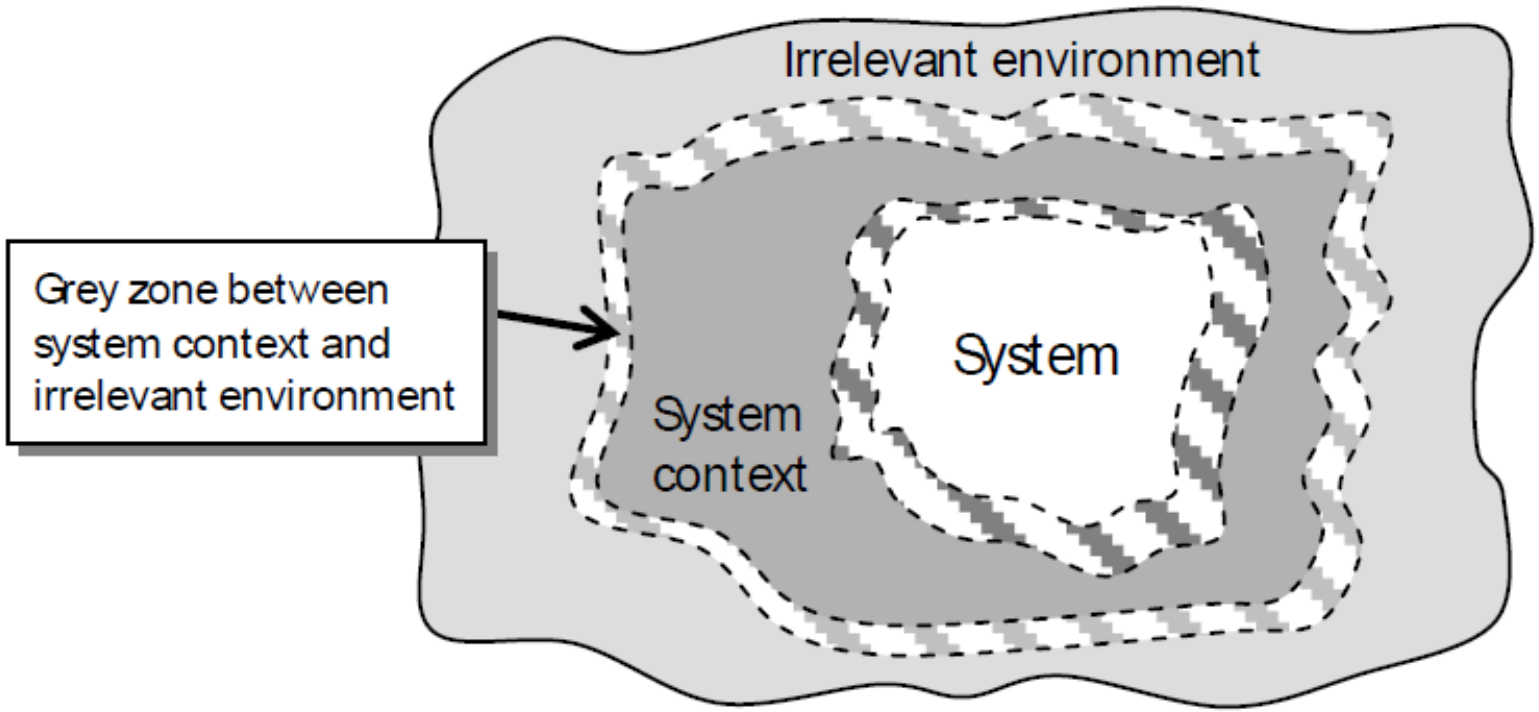
[Pohl 2010] « Requirements Engineering » §5.2

Limits of the system context

- Boundary between the relevant and the irrelevant part of the environment
- Its definition changes during the project
- Only the relevant part is taken into account



Yet another grey zone



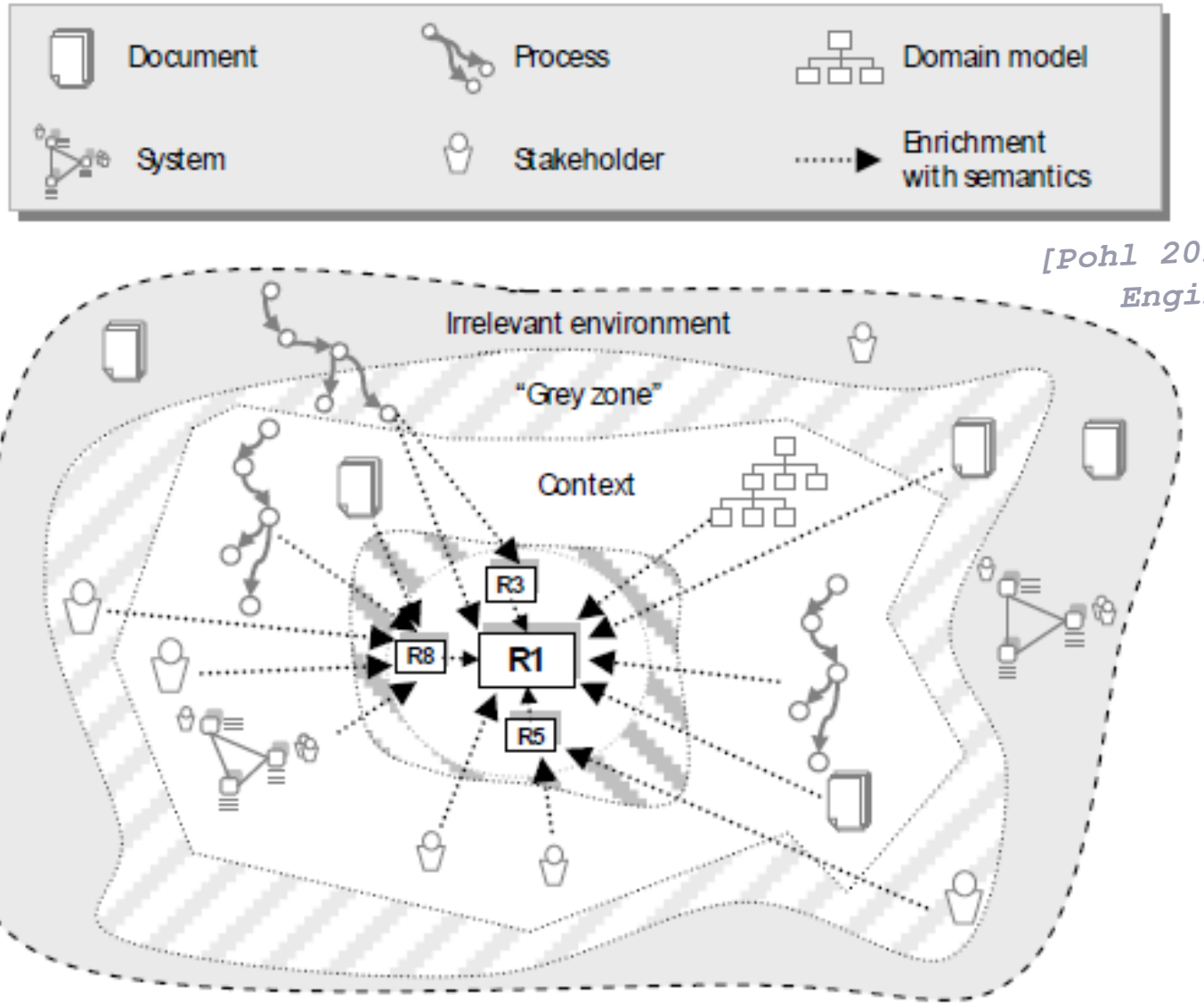
[Pohl 2010] « Requirements Engineering » §5.3

Document the context

- Requirements depend on the context
  - Domain knowledge
  - business rules, business processes, the organisation, the corporates strategy, etc
  - Law, norms, standards, etc
  - Other systems, technology,
- **Be careful, they can be assumptions about the future !**

[Pohl 2010]
« Requirements Engineering »
§5.4

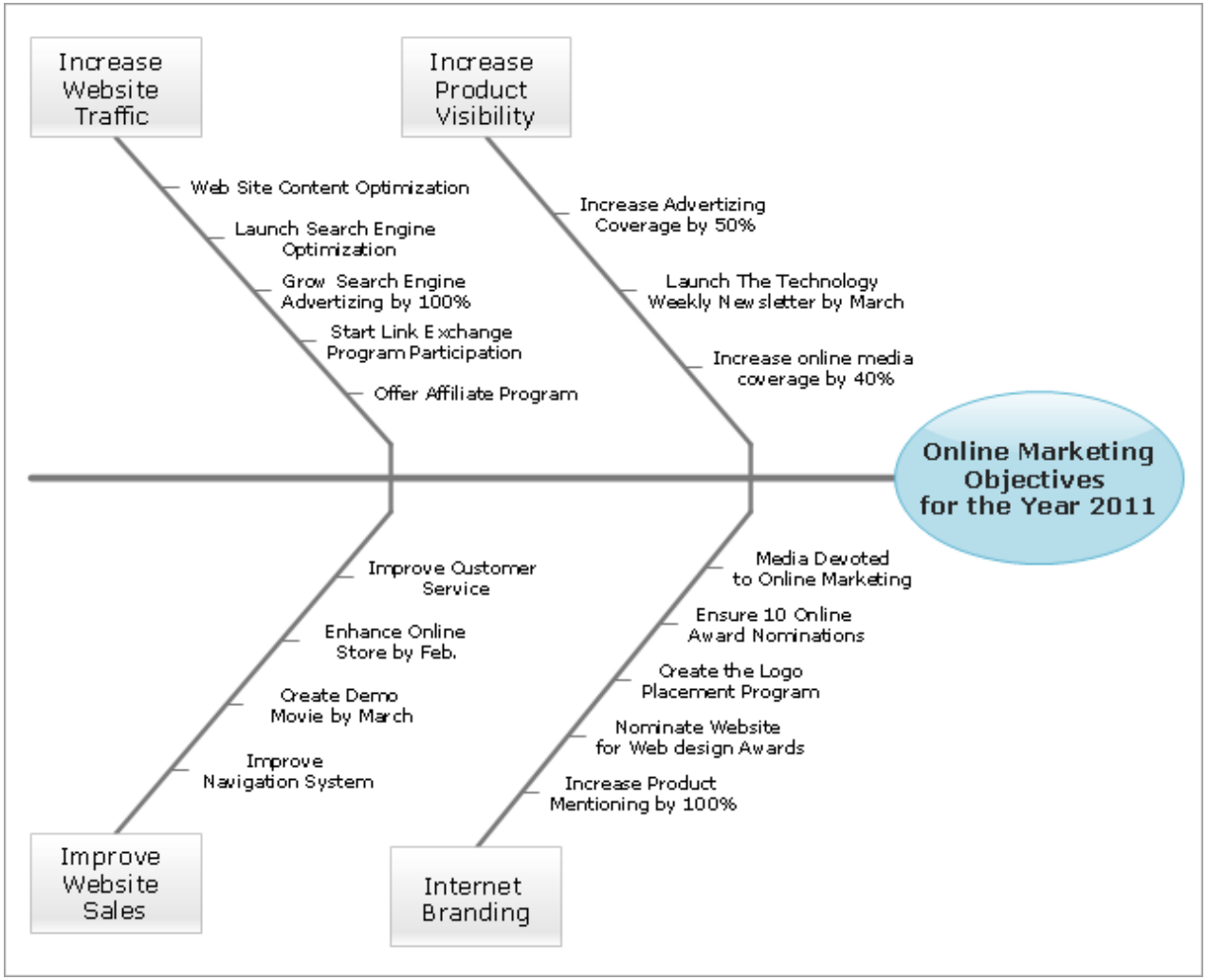
The context of a requirement



[Pohl 2010] « Requirements Engineering » §5.4

Good practice (1)

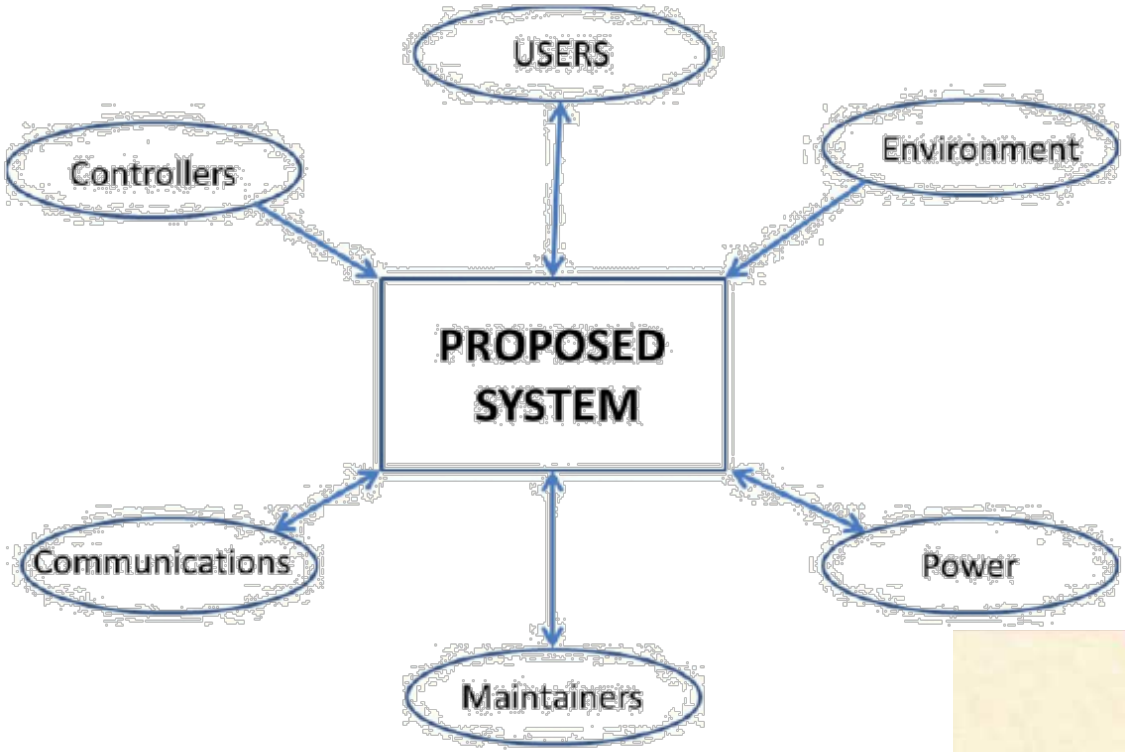
Strategic goals: Ishikawa's diagram

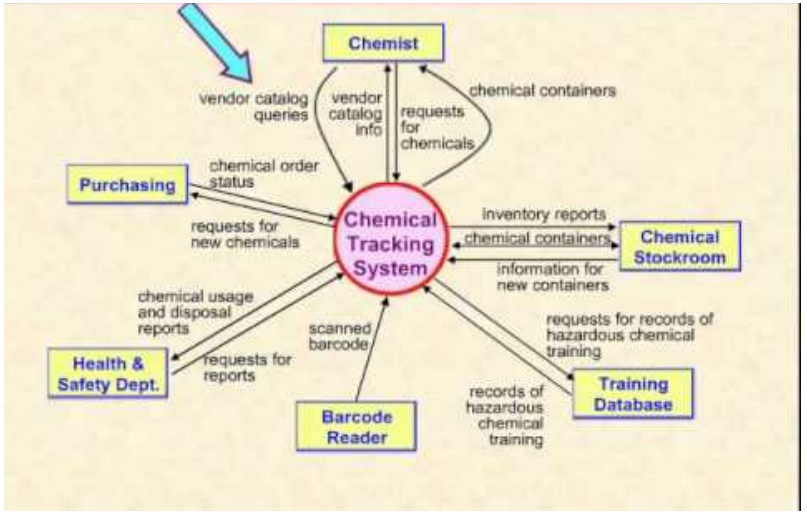


Source : « strategy-management-diagrams » http://www.conceptdraw.com/

Good practice (2)

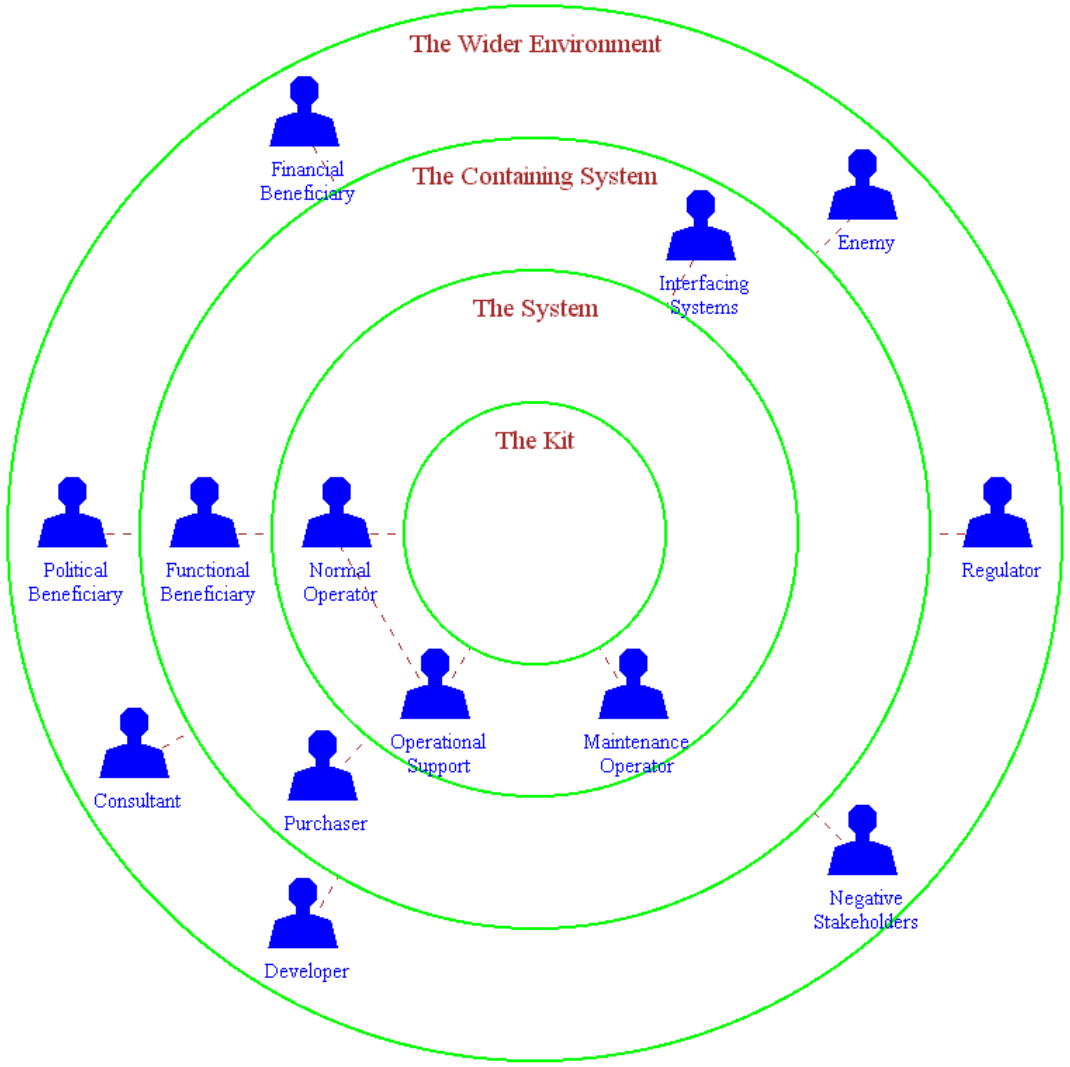
Frontiers of the system
Context diagrams



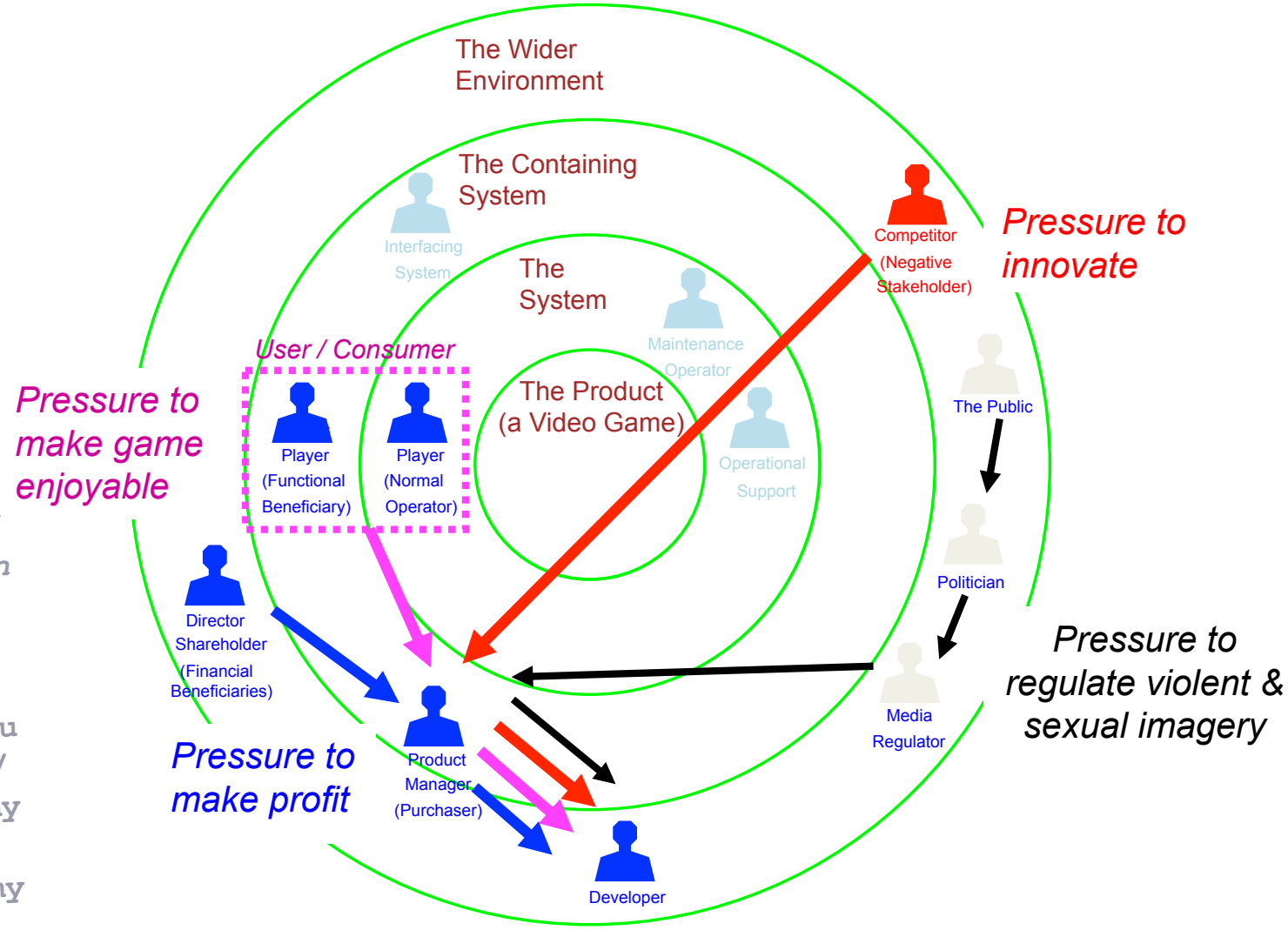


Good practice (3)

Stakeholders « stakeholder diagram »



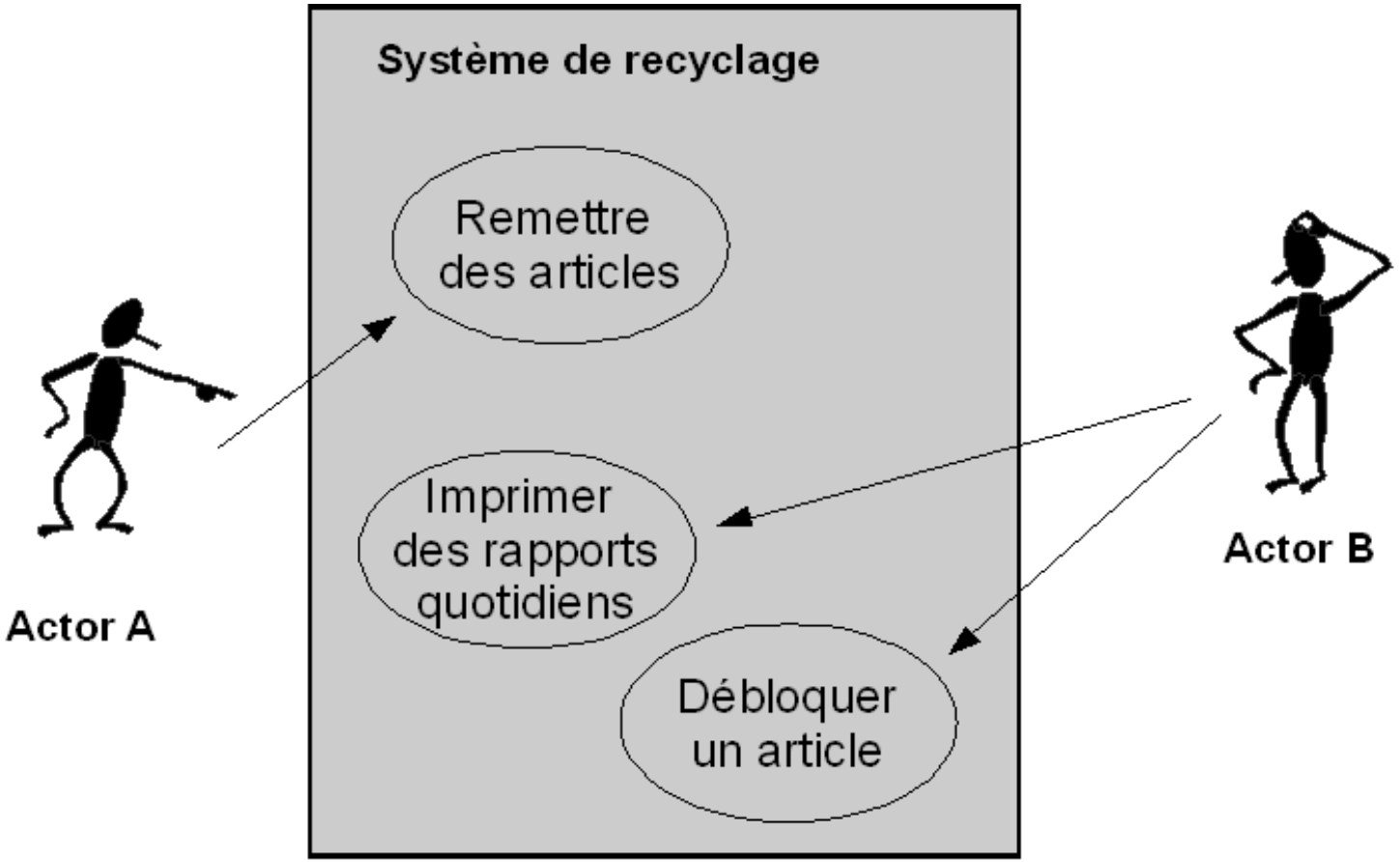
Example : influences amongt stakeholders in a videogame project



Ian F. Alexander 2005 "A Taxonomy of Stakeholders - Human Roles in System Development" http://easyweb.easynet.co.uk/iany/consultancy/stakeholder_taxonomy/stakeholder_taxonomy.htm

Good practice (4)

Usages
Use Case Diagrams



Use Case Diagrams

Do

- Identify actors
- Identify their goals
- Reify (services provided by the system)
- Complementary cases (nominal)

Don’t

- CRUD
- « Manage »
- Mix up levels of details
- Alternative cases (non nominal cases, exceptions, etc)

Exercise

- What is the context?
- Who are the stakeholders ?
- What are the main use cases?



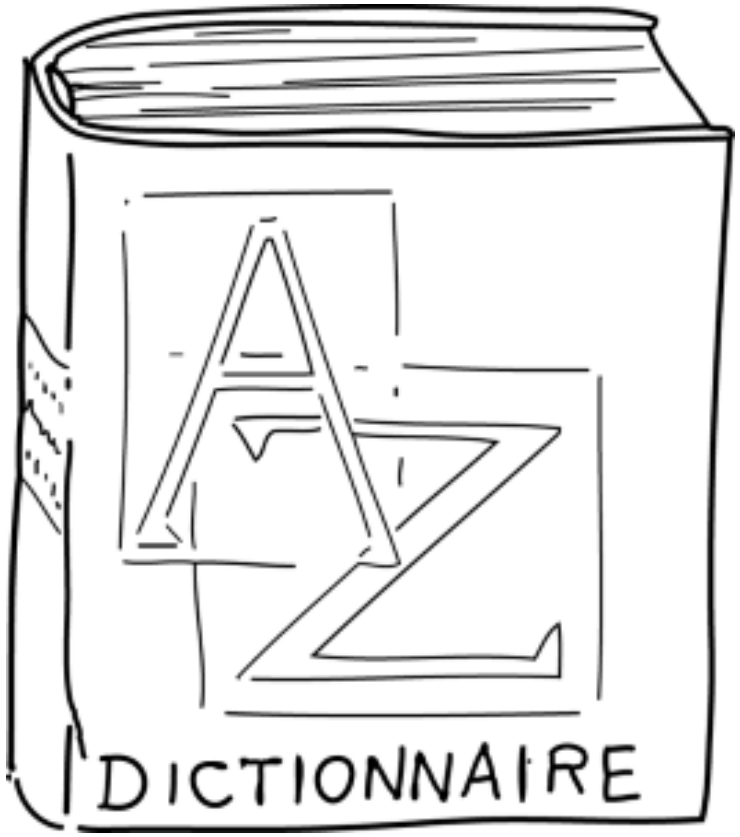
Good practice (5)

Don’t forget

- Domain knowledge
- Tacit knowledge and implicit requirements
- Law, norms, standards
- Constraints
- Context evolution
- Project requirements
- Associated systems
- Orthogonal requirements
- Interfaces with other systems and interoperability issues
- Legacy integration issues

- Assumptions about:
  - Market
  - Physical environment
  - Business environement (organiation, process, rules, etc)
  - Users (skills, expectations, needs…)

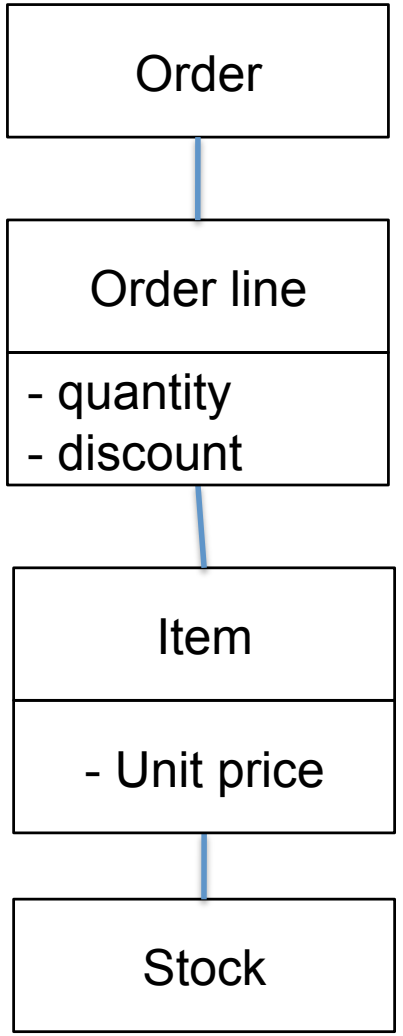
Domain dictionary



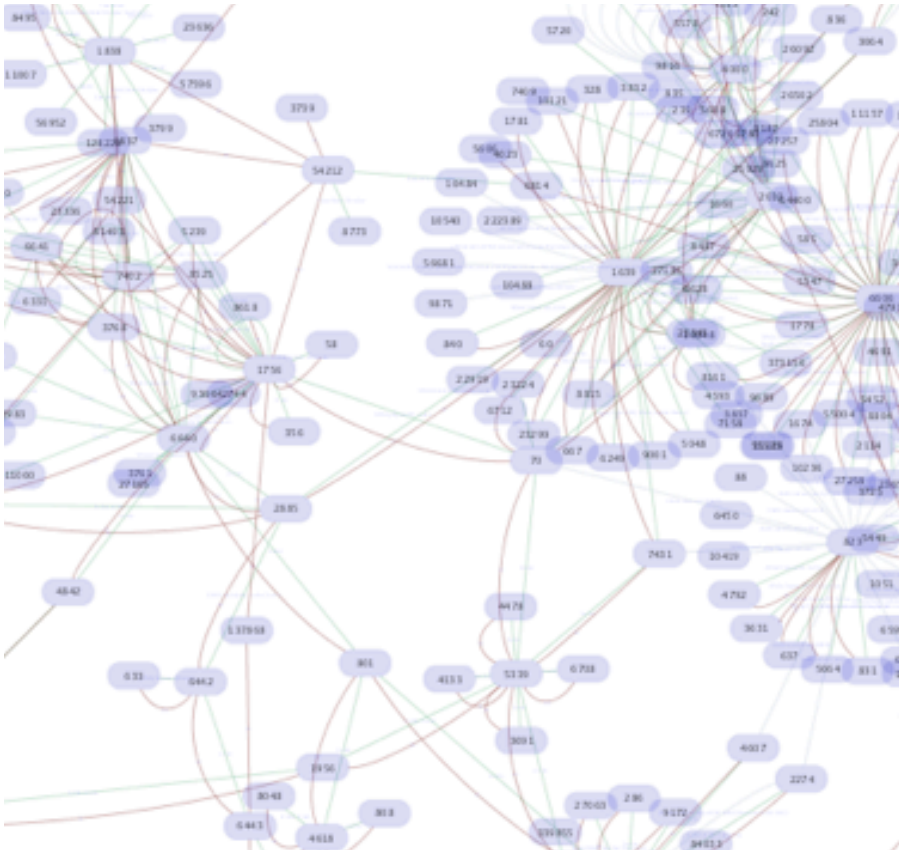
- Terms: actors, ressources, tasks information, rules, etc
- Acronyms, abreviations
- Definitions
- References/Sources
- Author

Domain models

- Formalized description of real world
- Shared by all
- Not implemented (abstract only)
- Uses a notation (E/R, BPMn, etc)



Ontologies



- Notations
  - Formal network of concepts
  - Logical rules
- Tools
  - Term extractors
  - Editors/modelers
  - Mapping tools
  - Solvers

Legacy analysis

Goal :

- Understand the current situation
- Exploit requirements and knowledge that are already documented

Means:

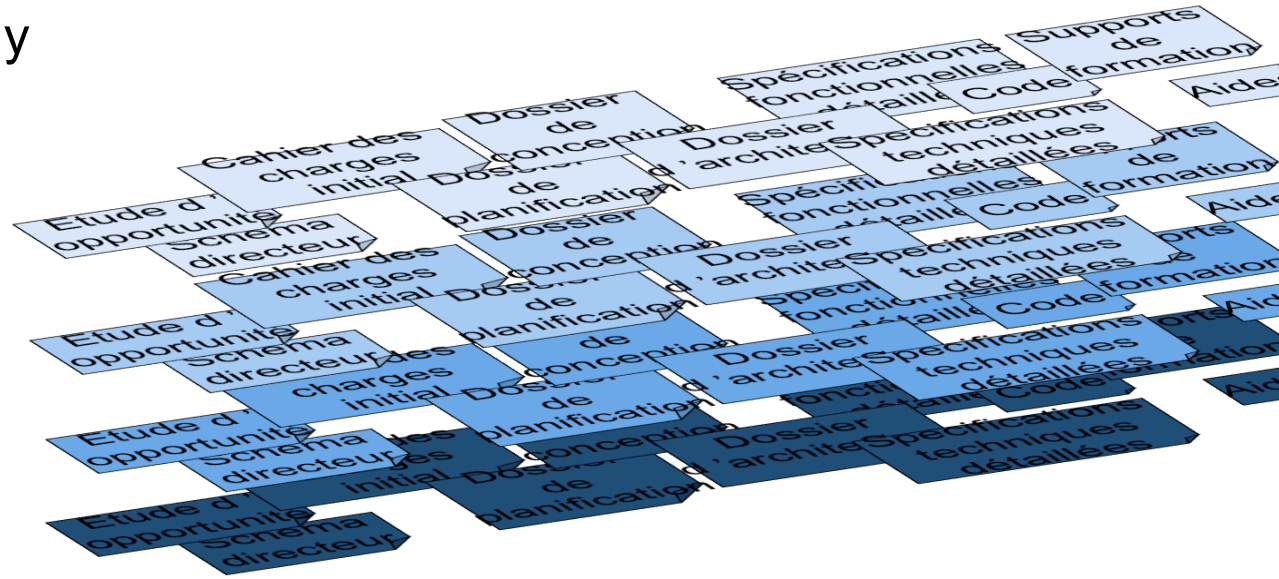
- Gather, evaluate, fitler information (plans, code, diagrams) or

Limitations :

- Needs critical view

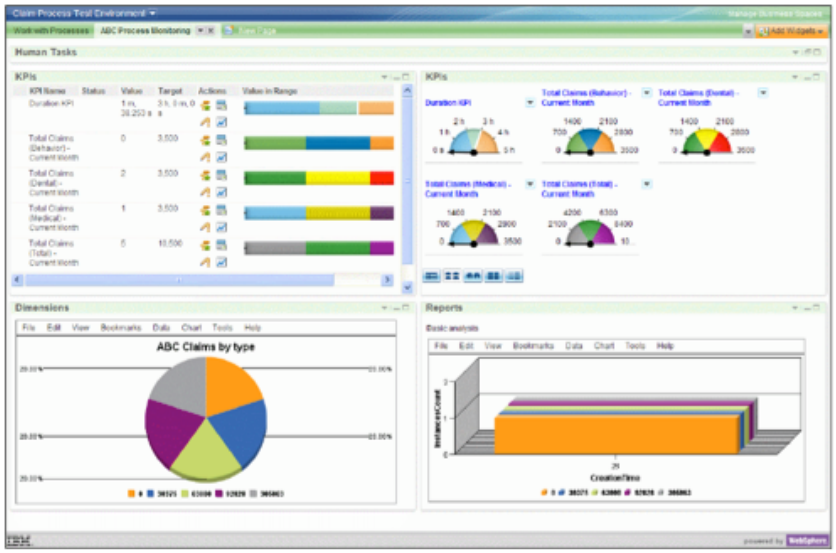
Practices:

- System archeology
- « perspective-based » reading
- Requirements reuse



Source : [Pohl-Rupp 2011] "Requirements Engineering Fundamentals – CPRE IREB"

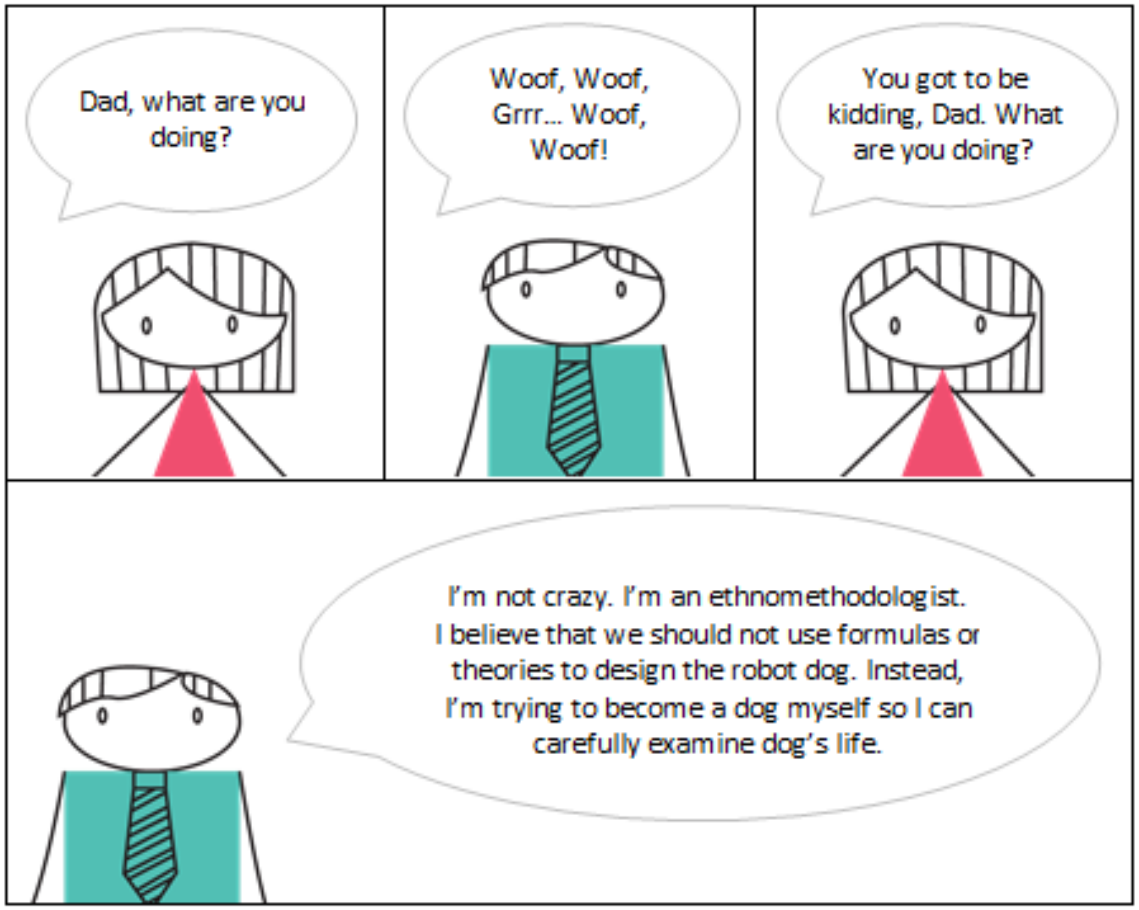
Cost and performance analysis



- Process execution time and frequencies
- Productivity issues, idle time, bottelnecks
- Latency, size of work piles, communication delays
- Error rates
- etc

Source: www.ibis-instruments.com

« quick and dirty » Ethnography

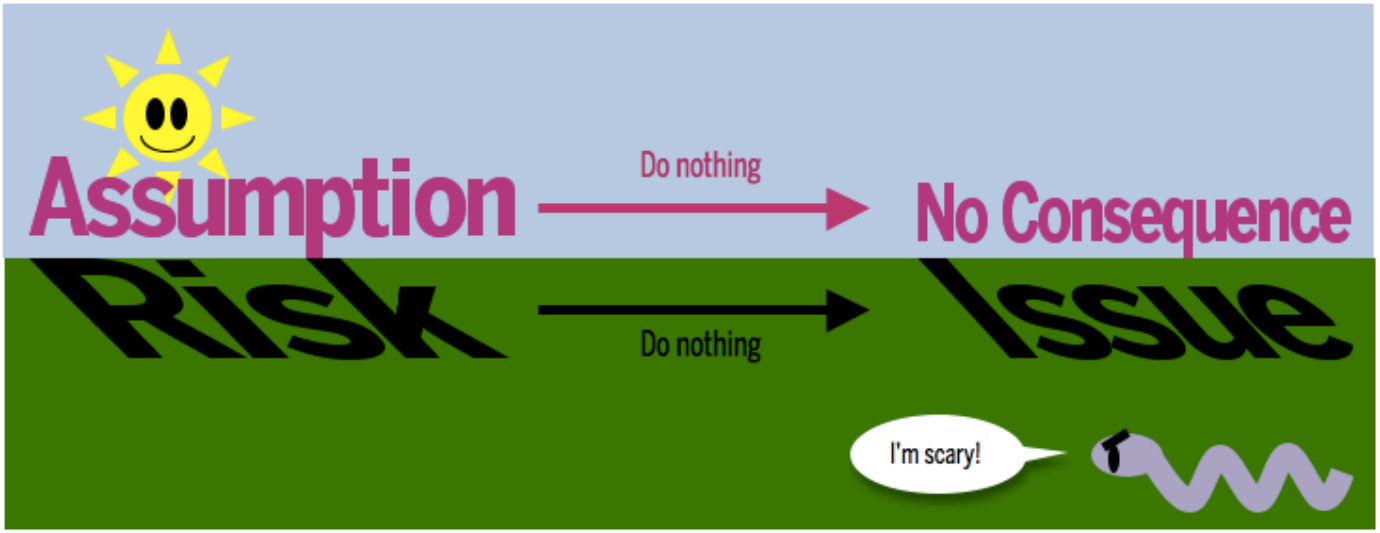


Source: theoryfamily.files.wordpress.com

Take away

- Start by understanding the context
  - Discuss it
  - Descibre it
  - Eliminate the “fat”
  - Foresee its evolutions
  - Look for its impacts on the requirements

Bad assumptions



… occur ***either***
because requirement authors **do not have access to sufficient information**
or
the information does not exist about:

- Needs
- Goals
- Objectives
- Constraints
- Missions
- Operations Concept
- Budget
- Schedule
- Management/Organization

Terms and definitions to remember

- System Context
- System Boundary
- Context Boundary